BRAIn lab
basic research of artificial intelligence

LECTURE 9

Retrieval-Augmented Generation

Documents, retrieval and grounded answers

Denis Shveykin

From documents to grounded answers

01 Corpus

Extraction, passages and search scope

02 Retrieval

Word matching and learned retrieval

03 RAG methods

Queries, retrieval control and context

04 Evaluation

Retrieval, answers and uncertainty

Prepare documents, search them, generate answers and evaluate the system.

The answer depends on the company's current policy

A CUSTOMER ASKS

"Can I return a customized desk after 10 days?"

Company facts may be private.

Policies change after model training.

The answer needs an inspectable source.

RETURNS POLICY · V2

Standard items: 30 days.

Customized items: no returns.

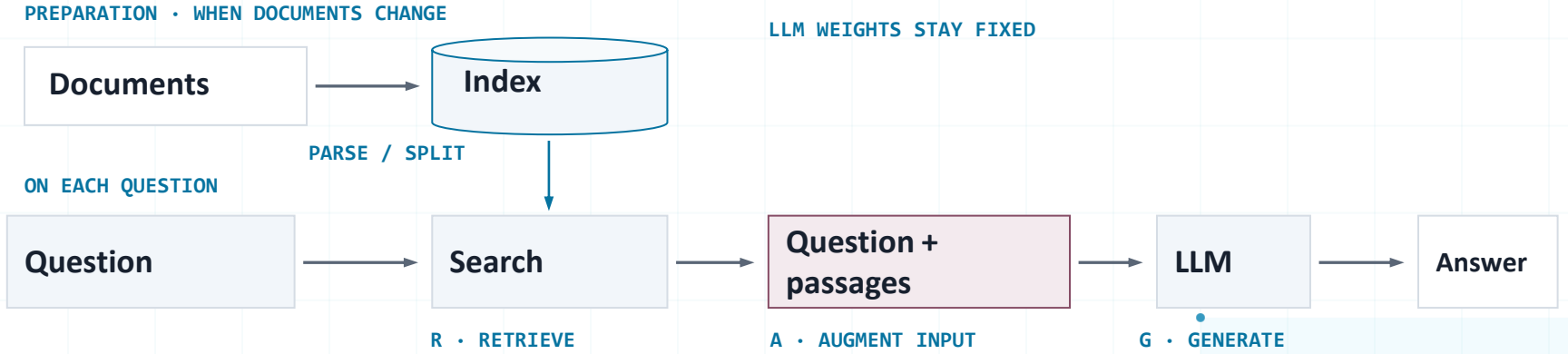
Internal document / section 4

Illustrative company policy. Assume this private document was never included in the LLM's training data.



RAG = Retrieval-Augmented Generation

The application finds relevant text and gives it to the LLM together with the question.



Search external documents, include relevant passages in the LLM's input, then generate an answer.

The retriever selects passages for this question

“Can I return a customized desk after 10 days?”

SEARCHABLE PASSAGES

4.1	**Standard items may be returned within 30 days.**
4.2	**Customized items cannot be returned.**
8.1	Delivery takes five business days.

RETURNED CONTEXT

Sections 4.1 and 4.2

The general rule and its exception are both available to the LLM.

Illustrative passages and retrieval result. The full corporate corpus can contain many other documents.



The LLM reads the policy inside its prompt

```
INSTRUCTION
Answer using the policy. Cite its section.

QUESTION
Can I return a customized desk after 10 days?

CONTEXT [returns-v2/4]
Standard items: returns within 30 days.
Customized items: no returns.
```

GENERATED ANSWER

No. Customized items cannot be returned. [returns-v2/4]

New facts enter the prompt.
LLM weights stay fixed.

Illustrative prompt and supported answer. The application supplies the passages and their source identifiers.



BRAIN lab
basic research of artificial intelligence

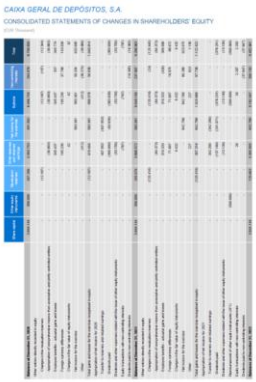
01

Preparing the corpus



What survives parsing, segmentation and indexing?

A PDF page can lose evidence before retrieval




```
{
  "text": ",QGLFDWHglyph<c=3,font=/LBKBHL+TimesNewRoman>E\\glyph<c=3,font=/LBKBHL+TimesNewRoman>FKHFNglyph<c=3,font=/
  LBKBHL+TimesNewRoman>PDUNglyph<c=3,font=/LBKBHL+TimesNewRoman>LIglyph<c=3,font=/LBKBHL+TimesNewRoman>WKHglyph<c=3,
  font=/LBKBHL+TimesNewRoman>5HJLVWUDQWglyph<c=3,font=/LBKBHL+TimesNewRoman>LVglyph<c=3,font=/LBKBHL
  +TimesNewRoman>Dglyph<c=3,font=/LBKBHL+TimesNewRoman>ZHOOglyph<c=16,font=/LBKBHL+TimesNewRoman>NQRZQglyph<c=3,font=/
  LBKBHL+TimesNewRoman>VHDVRQHGglyph<c=3,font=/LBKBHL+TimesNewRoman>LVVXHU glyph<c=15,font=/LBKBHL
  +TimesNewRoman>glyph<c=3,font=/LBKBHL+TimesNewRoman>DVglyph<c=3,font=/LBKBHL+TimesNewRoman>GHILQHGglyph<c=3,font=/
  LBKBHL+TimesNewRoman>LQglyph<c=3,font=/LBKBHL+TimesNewRoman>5XOHglyph<c=3,font=/LBKBHL+TimesNewRoman>glyph<c=23,
  font=/LBKBHL+TimesNewRoman>glyph<c=19,font=/LBKBHL+TimesNewRoman>glyph<c=24,font=/LBKBHL+TimesNewRoman>glyph<c=3,
  font=/LBKBHL+TimesNewRoman>RIglyph<c=3,font=/LBKBHL+TimesNewRoman>WKHglyph<c=3,font=/LBKBHL
  +TimesNewRoman>GHFXULWLHVglyph<c=3,font=/LBKBHL+TimesNewRoman> $FWglyph<c=17,font=/LBKBHL+TimesNewRoman>glyph<c=3,
  font=/LBKBHL+TimesNewRoman> <HV glyph<c=3,font=/LBKBHL+TimesNewRoman>1R glyph<c=0,font=/LBKBKM+Arial> glyph<c=0,
  font=/LBKBHM+Arial,Bold>",
  "type": "text",
  "text_id": 19
},
```


A readable page and an extractable text layer are different properties.

Original ERC2 parsing examples: rotated tables and corrupted font encodings.

Turn a table row into a self-contained passage

“What was MOL’s shareholders’ equity in 2021/3?”

SOURCE TABLE · AMOUNTS IN JPY MILLIONS

Metric	2020/3	2021/3	2022/3
Shareholders’ equity	513,335	577,782	1,274,570

PASSAGE FOR THE SEARCH INDEX

In 2021/3, MOL’s shareholders’ equity was JPY 577,782 million.

MOL values from the ERC2 retrospective, redrawn. The indexed sentence is an illustrative representation.

Chunking creates units for search

SOURCE DOCUMENT

Returns policy

Standard items
Customized items
Damaged items

INDEXED PASSAGES

01 Standard items: 30-day returns.

02 **Customized items: no returns.**

03 Damaged items: contact support.

Question: Can I return a customized item? → Retrieve passage 02

Split a document into passages, index each passage, and retrieve the relevant one.



Find a passage, read the full section

1 · RETRIEVED PASSAGE

Customized items cannot be returned.

The search finds a short matching passage.

2 · SECTION SENT TO THE MODEL

Standard items: 30-day returns.

Customized items: no returns.

Damaged items: contact support.

Search small passages and give the model their surrounding context.

Different search keys can identify the same evidence

ORIGINAL PASSAGE

Unused leave may be carried over until the end of the following quarter.

POSSIBLE SEARCH KEYS

SUMMARY

Leave carry-over and expiry

QUESTION

When does unused leave expire?

METADATA

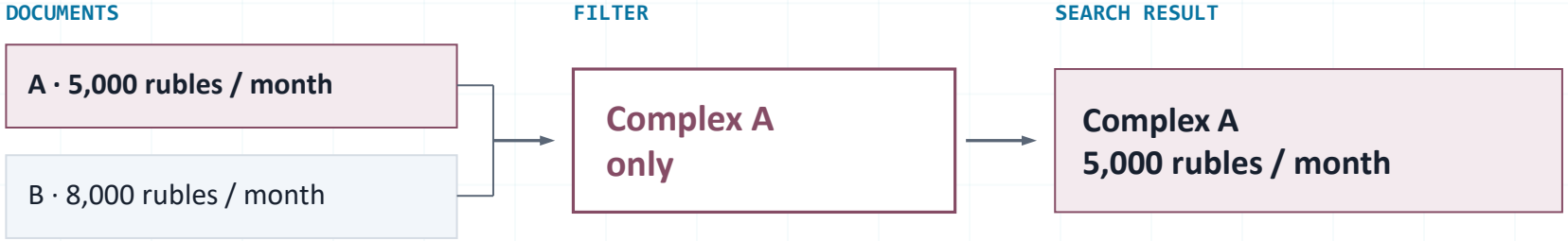
policy: leave / section: carry-over

Illustrative multi-representation indexing. The answer still comes from the original passage.

Limit the search to the right complex

QUESTION

“How much is parking at Complex A?”



The filter excludes Complex B before searching.

Illustrative example: the documents and prices are fictional.

BRAIN lab
basic research of artificial intelligence

02

Finding the evidence



Relevance, supervision and the cost of interaction

BM25: word matches, rarity and length

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t) \frac{f(t,d)(k_1+1)}{f(t,d) + k_1(1 - b + b|d|/\mathrm{avgdl})}$$

q: query · d: passage · t: query term

RARITY · INVERSE DOCUMENT FREQUENCY

$$\mathrm{IDF}(t) = \ln\left(1 + \frac{N - n_t + 0.5}{n_t + 0.5}\right)$$

N: number of indexed passages
n_t: passages containing term t

Rarer terms receive a larger weight.

FREQUENCY AND LENGTH

f(t,d): term count in the passage
|d|: passage length
avgdl: average passage length

k1: controls frequency saturation
b: controls length normalization
b = 0: off; b = 1: full

Sum a contribution for each query term; a missing term contributes zero.

DPR learns a ranking from positive and negative passages

$$s(q,p) = E_Q(q)^\top E_P(p)$$

$$\mathcal{L} = -\log \frac{\exp(s(q,p^+))}{\exp(s(q,p^+)) + \sum_j \exp(s(q,p_j^-))}$$

Training role	Illustrative query–passage pair
Query	When does the warranty expire for ZX-410?
Positive	ZX-410 warranty ends 24 months after purchase.
Hard negative	ZX-400 warranty ends 36 months after purchase.

Dense Passage Retrieval (DPR) learns separate question and passage vectors.

The negative sampling scheme changes retrieval accuracy

Type	#N	IB	Top-5	Top-20	Top-100
Random	7	✗	47.0	64.3	77.8
BM25	7	✗	50.0	63.3	74.8
Gold	7	✗	42.6	63.1	78.3
Gold	7	✓	51.1	69.1	80.8
Gold	31	✓	52.1	70.8	82.1
Gold	127	✓	55.8	73.0	83.1
G.+BM25$^{(1)}$	31+32	✓	65.0	77.3	84.4
G.+BM25$^{(2)}$	31+64	✓	64.5	76.4	84.0
G.+BM25$^{(1)}$	127+128	✓	**65.8**	**78.0**	**84.9**

READ TOP-20

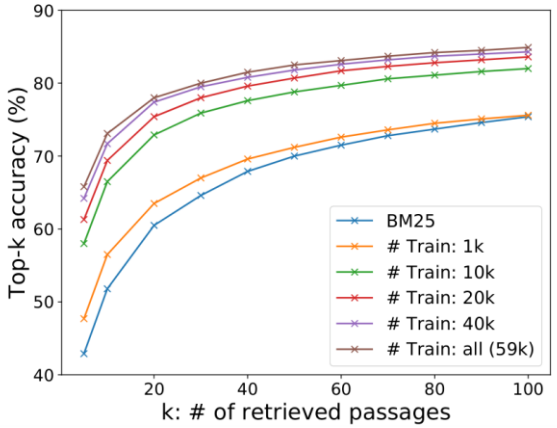
63.1 → 69.1

Seven labeled negatives per query: without vs. with in-batch reuse.

In-batch: use other queries' passages as extra negatives.

Natural Questions: percentage of queries with an answer-bearing passage in the top 20.

Dense retrieval depends on the supervision volume



HORIZONTAL AXIS

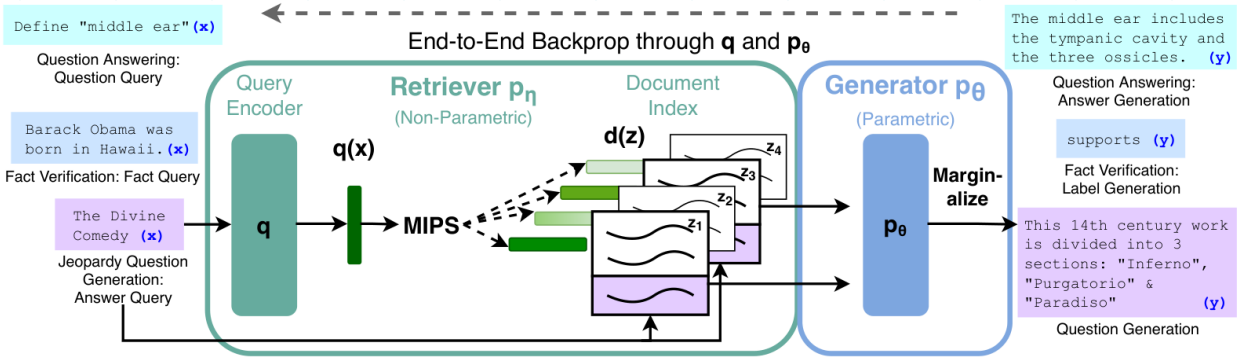
k: passages returned per query.

CURVES

1k to 59k training examples.
BM25 is the non-learned reference.
More retrieved passages improve coverage.

DPR Figure 1. Natural Questions development set; each curve uses a different number of training examples.

RAG trained retrieval and generation together



A query selects documents; the generator conditions on them. The original recipe also optimizes the query encoder.

Original RAG: the query encoder and generator are trained together.

RRF merges ranks rather than raw retrieval scores

$$\mathrm{RRF}(d) = \sum_{r\,:\,d \in r} \frac{1}{c + \mathrm{rank}_r(d)}$$

Document	BM25 rank	Dense rank	Fused score
D1	1	absent	0.01639
D2	2	1	0.03252
D3	absent	2	0.01613

D2 leads because both retrievers returned it, even though it was second in BM25.

Reciprocal Rank Fusion (RRF) combines ranked lists; c=60 in this example.

Reranking reads each candidate with the question

“What is the warranty period for ZX-410?”

Corpus → fast retrieval → top-k candidates → cross-encoder → top-m for the LLM

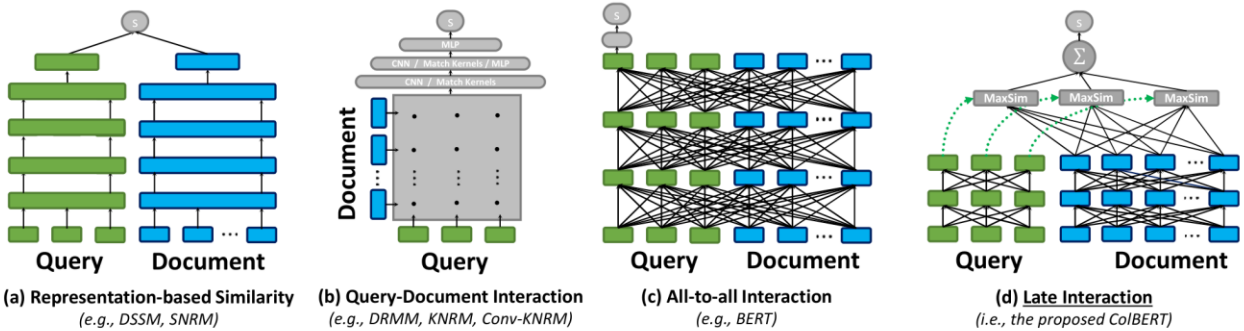
The cross-encoder jointly reads the question and one passage to produce a relevance score.

Before	Retrieved passage	After
1	D1 · Register your ZX-410 warranty online.	3
2	D2 · ZX-400 has a 36-month warranty.	2
3	D3 · ZX-410 has a 24-month warranty.	1

D3 moves to first place: both the product and the requested field match.

Illustrative ranking. The relevant passage must already be in the retrieved candidate set.

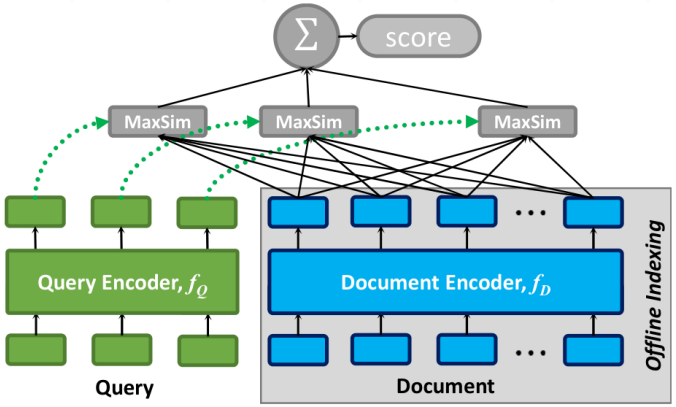
Where query and document are allowed to interact



ColBERT stores token vectors and compares them with the query at search time.

Bi-encoders compare separate vectors; cross-encoders jointly read each pair.

ColBERT scores the best document match for each query token



$$s(q,d) = \sum_i \max_j q_i^\top d_j$$

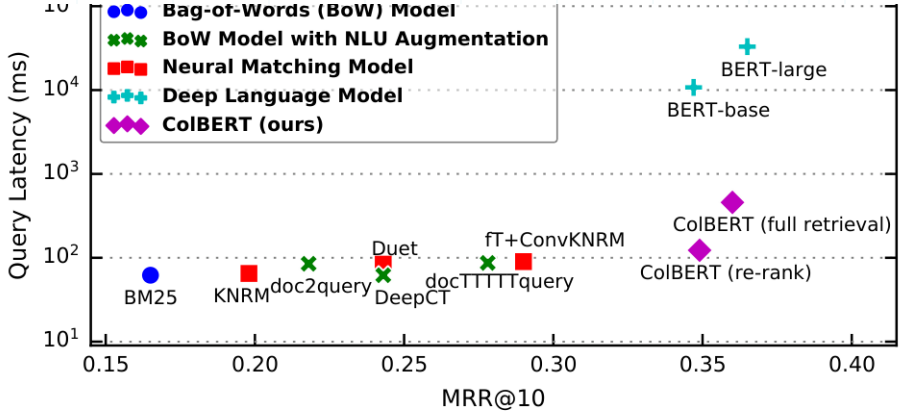
QUERY TOKENS × DOCUMENT TOKENS

0.9	0.2	0.1
0.1	0.8	0.2
0.2	0.3	0.7

0.9 + 0.8 + 0.7 = 2.4

Original Figure 3 and an illustrative normalized token-similarity matrix.

Interaction changes the quality–latency frontier



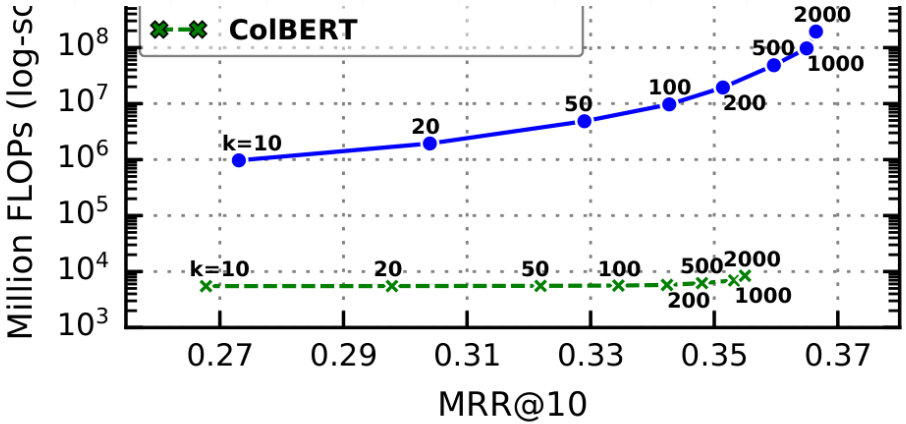
COMPARE BOTH AXES

Compare ranking quality with query latency on a log scale.

Reranking and full retrieval are different operating modes.

MS MARCO: MRR@10 averages the reciprocal rank of the first relevant result.

Reranking depth spends compute on candidate coverage



RERANKING DEPTH

More candidates increase coverage and pair-scoring work.

Reranking cannot recover a passage outside its candidate pool.

ColBERT Figure 4: k is the number of reranked passages; FLOPs measure computation.

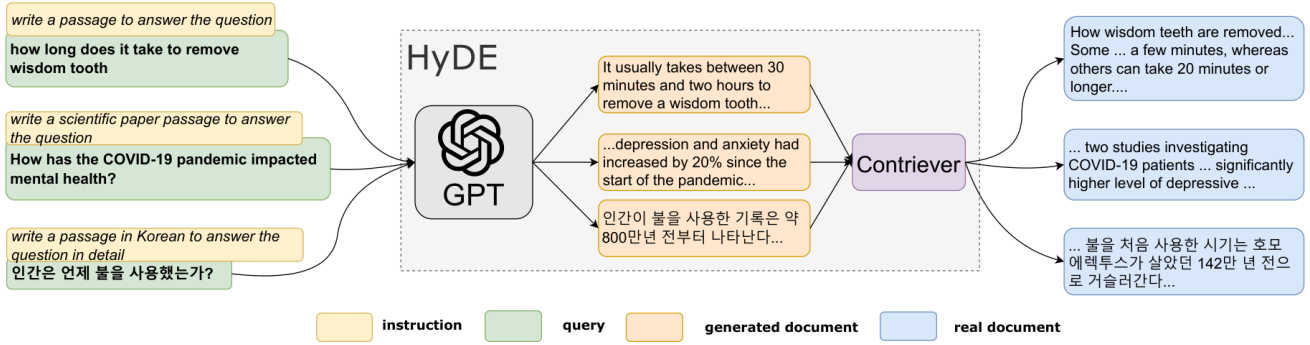
BRAIN lab
basic research of artificial intelligence

03

RAG methods and context

Query representation, retrieval control and global evidence

HyDE turns the query into a document-shaped key



The generated text may contain false details. Only retrieved source documents can support the answer.

Original HyDE Figure 1. The generated hypothetical document guides retrieval of real documents.

HyDE's gains depend on the retrieval setting

Retriever	SciFact	ArguAna	TREC-Covid	FiQA
BM25	67.9	39.7	59.5	23.6
Contriever (dense)	64.9	37.9	27.3	24.5
HyDE	69.1	46.6	59.3	27.3

TREC-Covid: a large gain over Contriever still does not beat BM25.

Retrieval tasks: nDCG@10 rewards relevant results ranked near the top (0–100).



Gao et al., 2023 — Precise Zero-Shot Dense Retrieval without Relevance Labels

Decompose a comparison into two searches

Which company had faster revenue growth from 2024 to 2025?

Search A: revenue in 2024 and 2025

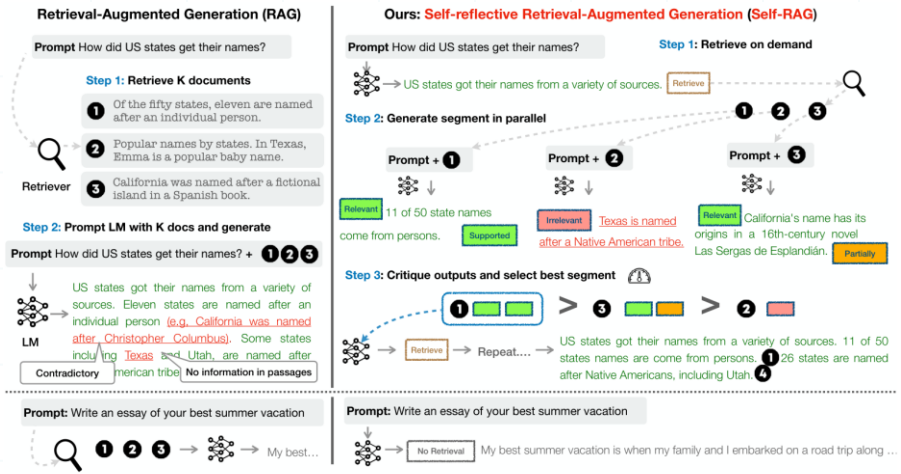
Search B: revenue in 2024 and 2025

Source	2024 revenue	2025 revenue	Computed growth
A / report-v2	USD 120 million	USD 150 million	25%
B / report-v1	EUR 80 million	EUR 100 million	25%

Compute each growth rate → compare: both companies grew by 25%.

Search each company’s reports, then compare the retrieved values. Illustrative data.

Self-RAG learns retrieval and critique decisions



Original Self-RAG Figure 1. Retrieval and reflection tokens belong to the trained model.

RETRIEVAL AND CRITIQUE

RETRIEVE

Obtain passages when needed

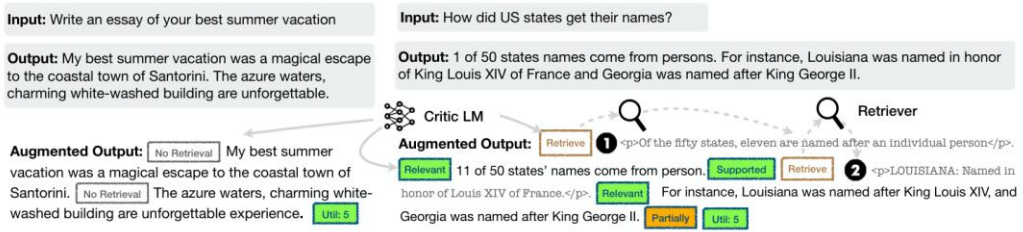
GENERATE

Continue using each passage

CRITIQUE

Score relevance, support and utility

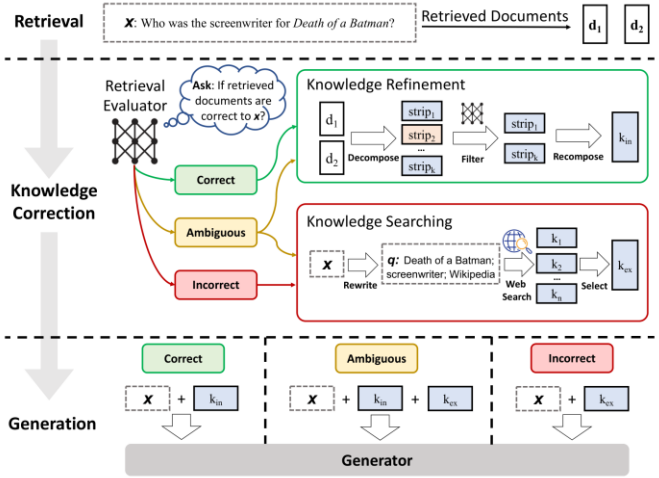
Reflection labels enter the training sequence



Signal	What the token labels
Retrieve	Whether retrieval is needed
IsRel	Passage relevance
IsSup	Support for the generated continuation
IsUse	Overall utility of the response

Original Self-RAG Figure 2. One example needs no retrieval; the other inserts retrieved passages and critiques.

CRAG chooses a correction from retrieval quality



RETRIEVAL ASSESSMENT

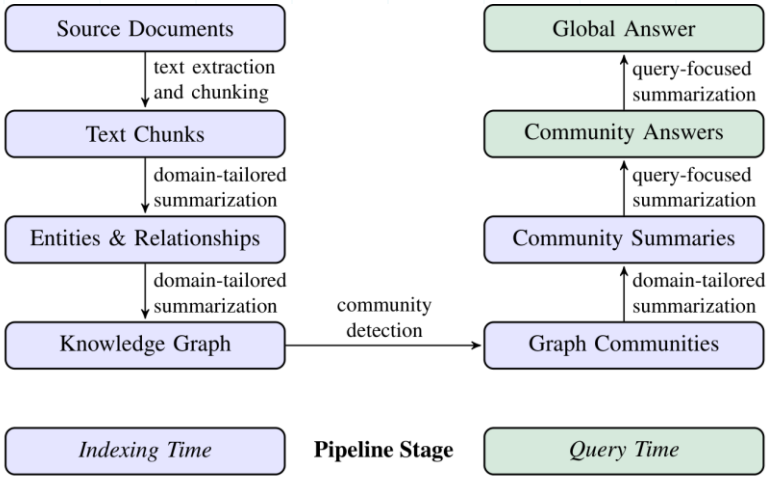
Correct: refine the retrieved documents.

Incorrect: seek external knowledge.

Ambiguous: combine the available knowledge paths.

Original CRAG Figure 2. Correct, ambiguous and incorrect retrieval trigger different knowledge paths.

GraphRAG changes the evidence for global questions



Indexing groups related entities into communities and summarizes each group.

A DIFFERENT QUESTION

“Which themes recur across the whole corpus?”

Local chunks may miss global coverage. Community reports summarize broader evidence.

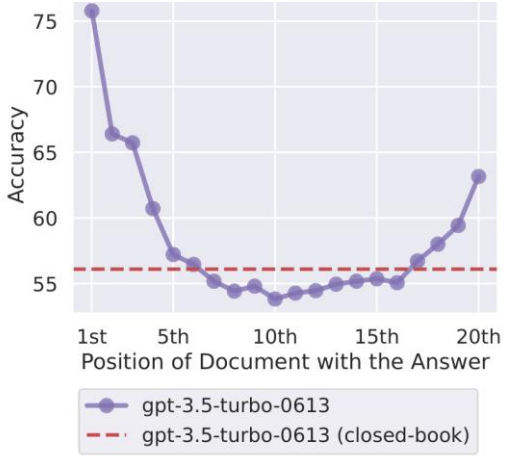
Global-answer evaluation rewards different properties

Comparison	Comprehensiveness	Diversity
C0 community reports vs. vector RAG	72% win rate	77% win rate

Inputs to summarization	Context units	Total context tokens
Top-level groups (C0)	34	26,657
Source text chunks (TS)	1,669	1,014,611

Podcast corpus: an LLM compares answers for coverage and diversity.

Context position can change the answer accuracy



VARIABLE: POSITION

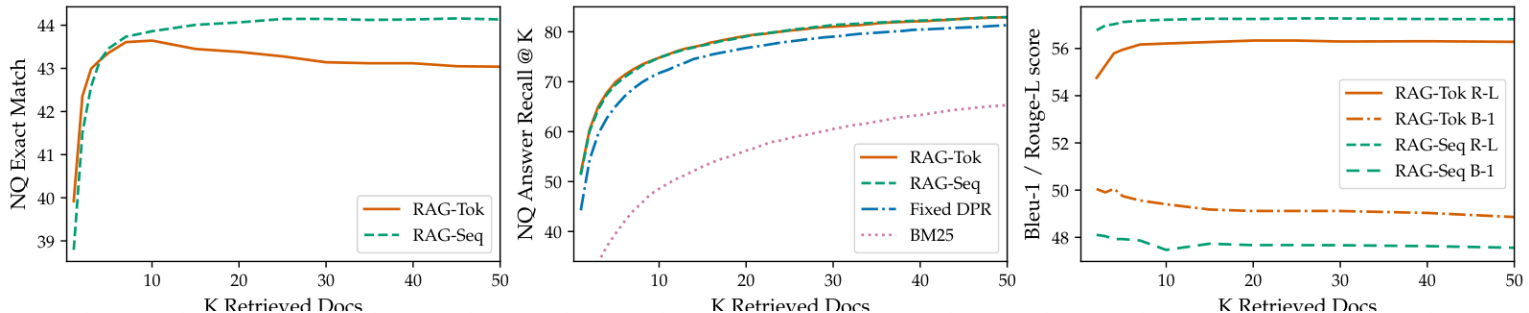
Only the position of the passage containing the answer.

FIXED CONDITIONS

The question, relevant evidence and total number of documents.

Lost in the Middle, Figure 1. Historical GPT-3.5-Turbo-0613, 20 retrieved documents, the same answer-bearing passage.

Retrieval recall and answer quality grow differently



RAG-Token answer EM peaks at 10 documents in this experiment, while retrieval recall keeps rising.

Natural Questions: EM is exact match; retrieval recall measures answer-containing passages.

Long context, retrieval and caching can be combined

Workload	A reasonable baseline	What to test
Small, static document set	Direct context + prompt caching	Cost and evidence utilization
Large, changing knowledge base	Search the updated index	Coverage and current information
Cross-document comparison	Retrieve all required sources, then read together	Entity, period and unit consistency

Choose a baseline based on corpus size, updates and the question.



Three ways to use a document corpus

Approach	What the model receives	Preparation
RAG	Retrieved text passages	Index the documents
Long context	The full corpus as text	Prefill the corpus
Cartridge	A compact learned KV cache	Train on the corpus offline

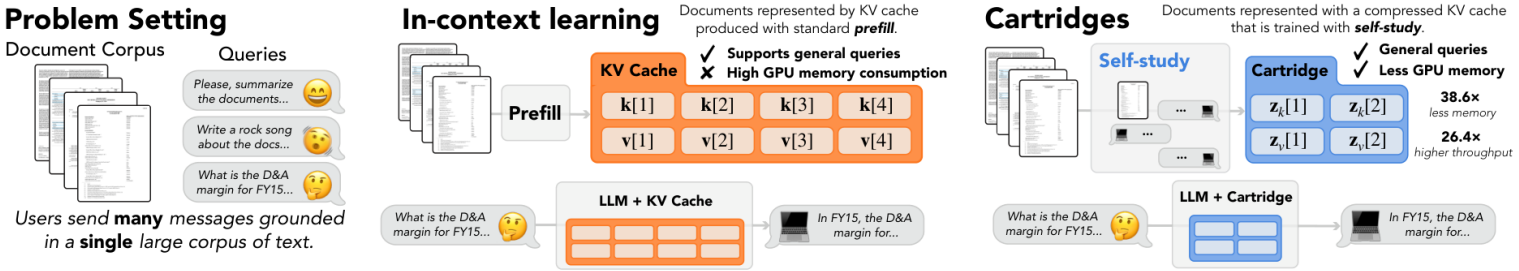
KV cache: stored attention keys and values that the model reads during generation.

A cartridge is a learned KV cache for repeated questions about the same corpus.



Eyuboglu et al., 2025 — Cartridges: Lightweight and general-purpose long context representations via self-study

Cartridges: train once, reuse across queries



OFFLINE · SELF-STUDY

Generate dialogues; train the cache to imitate reading the corpus.

INFERENCE · REUSE

Load the trained cache and append the user's question.

Original Figure 1. Self-study trains a compact KV cache while the LLM weights stay frozen.

BRAIN lab
basic research of artificial intelligence

04

Measuring the system

Retrieval coverage, supported claims and diagnostic interventions

Which retrieval failure does each metric expose?

Relevance by rank	Recall@3	Reciprocal rank	nDCG@3
0 → 1 → 1	2/3 = 0.667	0.500	0.531
1 → 0 → 1	2/3 = 0.667	1.000	0.704

Recall: documents found · RR: 1 / first relevant rank · nDCG: quality of the order

$$\mathrm{DCG@}k = \sum_{i=1}^{k} \frac{2^{rel_i} - 1}{\log_2(i+1)}$$

Three relevant documents exist; each row shows relevance at ranks 1–3 (1 = relevant).

Ragas evaluates retrieval and answers

Ragas is a framework for automated evaluation of RAG systems.

Inputs: question + retrieved passages + answer (+ reference when required)

LLM-based checks produce separate scores for different aspects of the system.

Stage	Metric	What it checks
Retrieval	Context precision	Are useful passages ranked above irrelevant ones?
Retrieval	Context recall	Do the passages cover facts needed for the reference answer?
Answer	Faithfulness	Are the answer's claims supported by the passages?
Answer	Answer relevance	Does the answer address the question?

Run the same test questions to compare changes to retrieval, chunking or generation.



Es et al., 2024 — RAGAS: Automated Evaluation of Retrieval Augmented Generation

Context precision and recall need different labels

Rank	Evidence verdict	Precision at this rank
1	Relevant	1 / 1
2	Irrelevant	1 / 2
3	Relevant	2 / 3

Context precision
(1 + 2/3) / 2 = 0.833

Context recall
2 supported facts / 3 = 0.667

Current Ragas collection	Inputs beyond the question
ContextPrecision / ContextRecall	retrieved_contexts + reference
ContextUtilization	retrieved_contexts + response

Manual verdicts for a teaching example, not a Ragas API run. The reference contains three required facts.



Ragas documentation — Available metrics and required inputs

A relevant response can still miss the question

Question	Answer
When is the scheduled launch date and time for the PSLV-C56 mission, and where will it be launched from?	*High answer relevance*: The PSLV-C56 mission is scheduled to be launched on Sunday, 30 July 2023 at 06:30 IST / 01:00 UTC. It will be launched from the Satish Dhawan Space Centre, Sriharikota, Andhra Pradesh, India. *Low answer relevance*: The scheduled launch date and time for the PSLV-C56 mission have not been provided.The PSLV-C56 mission is an important space mission for India. It aims to launch a satellite into orbit to study weather patterns.

Response relevance asks whether the answer addresses the question. It does not establish factual correctness.

Original RAGAS Table 3 (WikiEval). PSLV-C56: the question asks for the exact launch date and time.

Oracle evidence separates retrieval and reader errors

QUESTION: CAN A CUSTOMIZED ITEM BE RETURNED?

Context condition	Evidence available	What the comparison diagnoses
Retrieved	General return rule; exception absent	End-to-end behavior
Oracle	Full section including the exception	Reader given sufficient evidence
No evidence	No policy text	Parametric answer / refusal baseline

Retrieved fails + oracle succeeds → inspect retrieval. Oracle also fails → inspect reading and the task contract.

Compare retrieved passages with a manually selected complete section.

Error analysis needs inspectable records

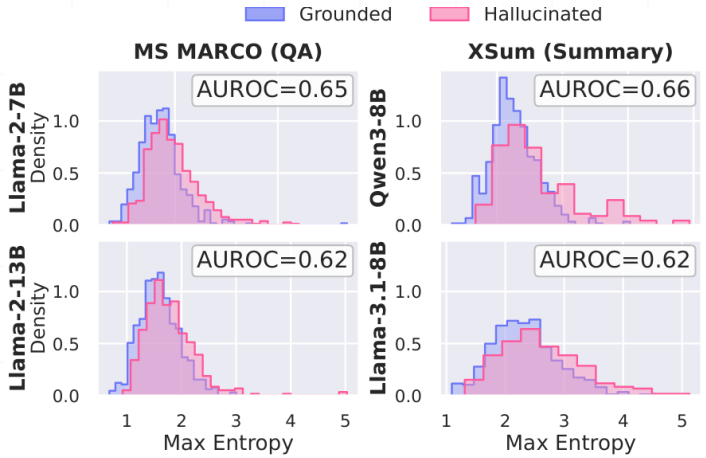
```
query: "Can a customized item be returned?"
retrieved: returns-v2 / section-4 / page-3
evidence: "Customized items cannot be returned."
answer: "Yes, within 30 days. [returns-v2/4]"
claim_support: false
citation_exists: true
error_stage: reader / ignored exception
```

Store source locations, revisions and judge verdicts; then inspect the actual failed examples.

Illustrative trace. A citation can point to a real document while the answer omits its decisive condition.



Entropy alone overlaps on grounded and false outputs



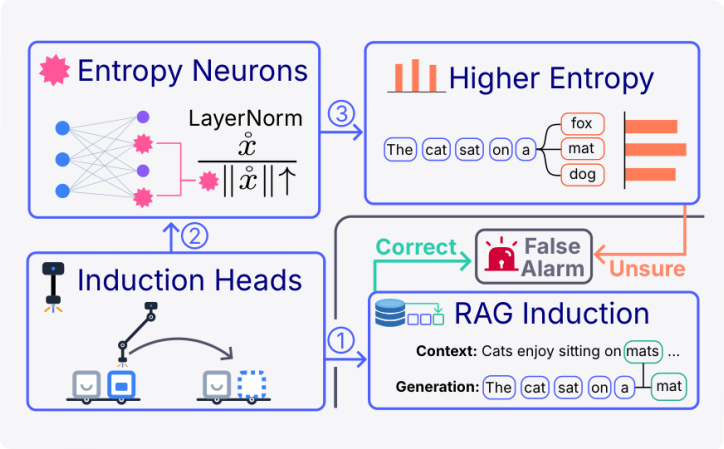
ENTROPY DISTRIBUTIONS

Grounded and hallucinated responses occupy overlapping ranges.

AUROC 0.62–0.66: how well the score ranks errors above grounded responses.

Entropy measures uncertainty; here the highest token entropy scores each response.

Induction has two opposing paths to entropy



DIRECT PATH

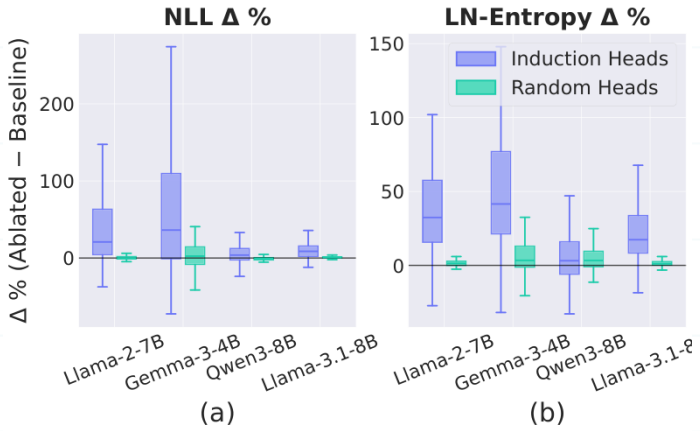
Induction supports the correct continuation and lowers entropy.

OPPOSING PATH

Co-activated entropy neurons enlarge the normalization denominator.

Induction helps reuse a pattern from context; it can also activate entropy neurons.

Ablation tests more than correlation



CONTROLLED INTERVENTION

Replace 5 induction heads with their mean output; compare with 5 random heads.

Loss and entropy rise more after induction-head ablation. This supports their role on this selected subset.

CoQA: correct responses with low SinkRate, a measure of attention concentration.

INTRYGUE gates entropy by induction activity

$$U(P,R) = f(s_1, \ldots, s_k) \cdot g(e_1, \ldots, e_n)$$

SINKRATE GATE

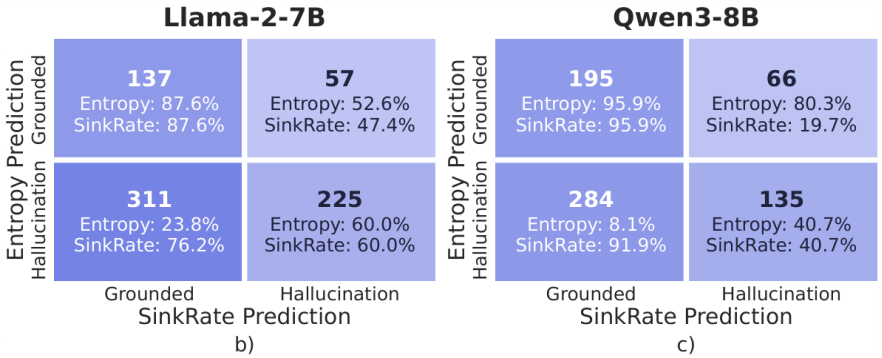
TOKEN ENTROPIES

SinkRate: attention concentrated on one position; low values indicate stronger induction.

Dataset / model	Baseline AUROC	INTRYGUE AUROC
MS MARCO / Mistral-7B	MaxEntropy: 0.67 ± 0.03	min-max: 0.77 ± 0.03
CoQA / Mistral-7B	LN-Entropy: 0.83 ± 0.03	mean: 0.82 ± 0.04

INTRYGUE Eq. 7 and Tables 1–2. AUROC mean ± deviation over five runs; aggregation matters.

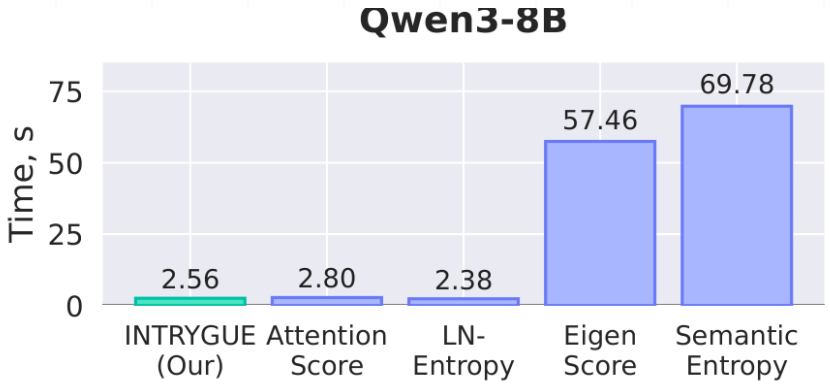
The two signals correct different mistakes



Bottom-left: entropy predicts a hallucination while SinkRate predicts a grounded answer.

INTRYGUE Figure 6(b–c), original panels, CoQA. Quadrants group the predictions of entropy and SinkRate.

Cost and limits of white-box scoring



DEPLOYMENT CONDITIONS

Requires model internals and attention capture.

No auxiliary model or extra generations for the score.

Grounding ≠ factual truth. The score does not secure a tool call.

INTRYGUE Figure 8. Qwen3-8B, 20 XSum samples, averaged over ten runs; values describe this setup.

BRAIN lab
basic research of artificial intelligence

LECTURE 9 · SYNTHESIS

Lecture 9 summary

Documents, retrieval and evaluation



Diagnosing errors in a RAG system

Observed failure	A targeted intervention	Check before declaring success
Fact absent from index	Repair extraction or segmentation	Compare source page and indexed text
Wrong entity or policy	Scope, metadata and routing	Test ambiguity and routing errors
Relevant fact ranked too low	Retrieval adaptation / reranking	Coverage and answer quality at fixed budget
Evidence present, answer wrong	Reader / prompt / training	Oracle evidence + claim support
Global question undercovered	Community reports or broader reading	Global coverage and summary fidelity
Grounded answer rejected	Calibrate the confidence signal	Error rate versus answered queries

Use the observed error to choose an intervention and evaluate it.



Reading and implementations

01 · FOUNDATIONS	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
02 · INTERACTION	ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT
03 · EVALUATION	RAGAS: Automated Evaluation of Retrieval Augmented Generation
04 · EMNLP 2026 ORAL	INTRYGUE: Induction-Aware Entropy Gating for Reliable RAG Uncertainty Estimation
05 · CASE + CODE	How I Won the Enterprise RAG Challenge
06 · COURSE MATERIAL	Retrieval Stuff

Full titles link directly to the paper or course material. Figure provenance and case sources are in the deck notes.



BRAIN lab
basic research of artificial intelligence

LECTURE 10

Agents

Actions, tools and feedback from the environment



Course repository

Choose an action → observe its result → revise the next action.